

ID6002W: Online and Reinforcement Learning

Course Syllabus for online learning (bandit) portion

Programme: Online M.Tech. in Data Science and AI (Elective), WSAI, IIT Madras

Duration: 7 Weeks (14 Lectures)

Prerequisites: Completion of Mathematical Foundations for Data Science or equivalent

Evaluation Strategy

- **Two Programming Assignments (25% each):** Focused on implementation and empirical analysis.
 - **Assignment 1:** Covering Lectures 1–5 (Due: End of Week 3).
 - **Assignment 2:** Covering Lectures 6–10 (Due: End of Week 6).
 - **One Comprehensive Final Examination (50%):** A 1.5 to 2-hour assessment covering the full module.
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Course Schedule

Week 1: Foundations of Sequential Decision-Making

- **Lecture 1: The Online Learning Paradigm**
 - Industry use cases: Recommendation systems, logistics matching, dynamic pricing, clinical trials, and ad selection.
 - Formalising the Multi-Armed Bandit (MAB): Arms, rewards, horizons, and the exploration–exploitation tradeoff.
 - The Bandit-vs-RL contrast: Sequential games (Chess/Go) vs. independent state assumptions in bandits.
 - Cumulative regret as the canonical objective for ongoing optimisation.
 - Simple strategies: The mechanics and production limitations of ϵ -greedy.
- **Lecture 2: Best-Arm Identification (BAI)**
 - BAI as adaptive A/B testing: Finding the best arm with high confidence.
 - Statistical foundations: Hoeffding's Inequality and concentration bounds.
 - Successive elimination algorithms and sample complexity.
 - Objective selection: When to prioritise BAI over regret minimisation.

Week 2: Stochastic Bandit Algorithms

- **Lecture 3: UCB Algorithms**
 - Explore-then-Commit (ETC) strategies and the transition from BAI.
 - The "Optimism in the Face of Uncertainty" principle.
 - Upper Confidence Bound (UCB1): Derivation, implementation, and regret bounds.
- **Lecture 4: Thompson Sampling (TS)**
 - Bayesian foundations: Beta–Binomial conjugacy and posterior updates.
 - Posterior sampling as a randomized exploration strategy.
 - Empirical comparisons: Performance and robustness to delayed/batched feedback.

Week 3: Practical Stochastic Bandits and Context

- **Lecture 5: Non-Stationarity and Practical Constraints**
 - Handling distribution shift: Sliding-window UCB and discounted Thompson Sampling.
 - Case Study: Spotify "Impatient Bandits" (Managing 60-day delayed rewards).
 - Reward design: Using progressive feedback and intermediate proxies for long-term engagement.
- **Lecture 6: Introduction to Contextual Bandits**
 - Incorporating observable context: User features, device, and temporal metadata.
 - Case Study: Yahoo! Front Page news personalisation.
 - Modern representations: Two-tower architectures and learned embeddings for context vectors.

Week 4: Linear Bandit Frameworks

- **Lecture 7: LinUCB**
 - Linear reward models and parameter estimation via ridge regression.
 - Geometric intuition: Confidence ellipsoids in high-dimensional feature spaces.
 - Regret analysis and the impact of feature dimensionality (d).
- **Lecture 8: Linear Thompson Sampling**
 - Bayesian linear regression: Multivariate Gaussian posteriors over weight vectors.
 - Posterior sampling for structured exploration in linear spaces.
 - Selection criteria: Comparative advantages of LinUCB vs. Linear TS in production.

Week 5: Evaluation and Production Systems

- **Lecture 9: Off-Policy Evaluation (OPE)**
 - The "Logging Trap": Requirements for context, action probabilities, and reward logging.
 - Inverse Propensity Scoring (IPS): Importance weighting and unbiased estimation.
 - Doubly Robust (DR) estimators: Combining reward models with IPS for variance reduction.
 - Production debiasing: Propensity clipping and self-normalised estimators.
- **Lecture 10: Production Architectures**
 - System design: Two-stage retrieval funnels (candidate generation) and ranking architectures.
 - Exploratory nominators and the impact of retrieval bias.
 - Case Study: Udemy's multi-armed bandit infrastructure for item ranking.

Week 6: Resource Constraints and Preference Feedback

- **Lecture 11: Bandits with Knapsacks (BwK)**
 - Online optimisation under global resource constraints (budget, inventory, or time).
 - LP relaxation and primal-dual algorithms for constrained bandits.
 - Applications: Dynamic pricing with finite inventory and ad budget pacing.
- **Lecture 12: Dueling Bandits**
 - Learning from ordinal feedback: Pairwise user preferences vs. numerical rewards.
 - Solution concepts: Condorcet winners and Borda/Copeland alternatives.
 - Algorithm families: Interleaved Filter, Relative UCB (RUCB), and Double TS.

Week 7: Structured Feedback and Synthesis

- **Lecture 13: Cascading Bandits**
 - The cascade click model: Modelling user interaction with ranked lists.
 - Handling position bias through structural assumptions and examination models.
 - CascadeUCB: Exploration over slates and ordered items.
- **Lecture 14: The Research Frontier and Bridge to RL**
 - Frontiers: Adversarial bandits (EXP3), neural bandits, and Bayesian optimisation.
 - Course synthesis: Transitioning from stateless bandits to state-evolving Reinforcement Learning.